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Quero

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(54) **WELL ABANDONMENT AND SEVERANCE
OF CONTROL LINES**

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E21B 33/12 (2006.01)

(52) **U.S. Cl.**
CPC **E21B 33/05** (2013.01); **E21B 33/1208**
(2013.01)

(58) **Field of Classification Search**
CPC E21B 33/05; E21B 33/1208
See application file for complete search history.

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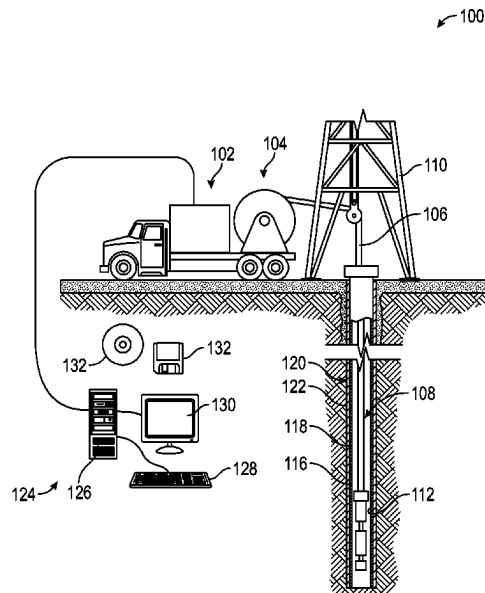
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Tumey Law Group PLLC

(57) **ABSTRACT**

A method may include injecting a sealing material into an annulus formed between a tubular and a wellbore wall. The sealing material is configured to surround at least a portion of a control line disposed in the annulus. Further, the method includes making at least one orbital cut, via a cutting device, through the tubular, the sealing material in the annulus, and the control line surrounded by the sealing material. Additionally, the method includes filling the at least one orbital cut and at least a portion of a central bore of the tubular with additional sealing material to seal the wellbore.

10 Claims, 7 Drawing Sheets



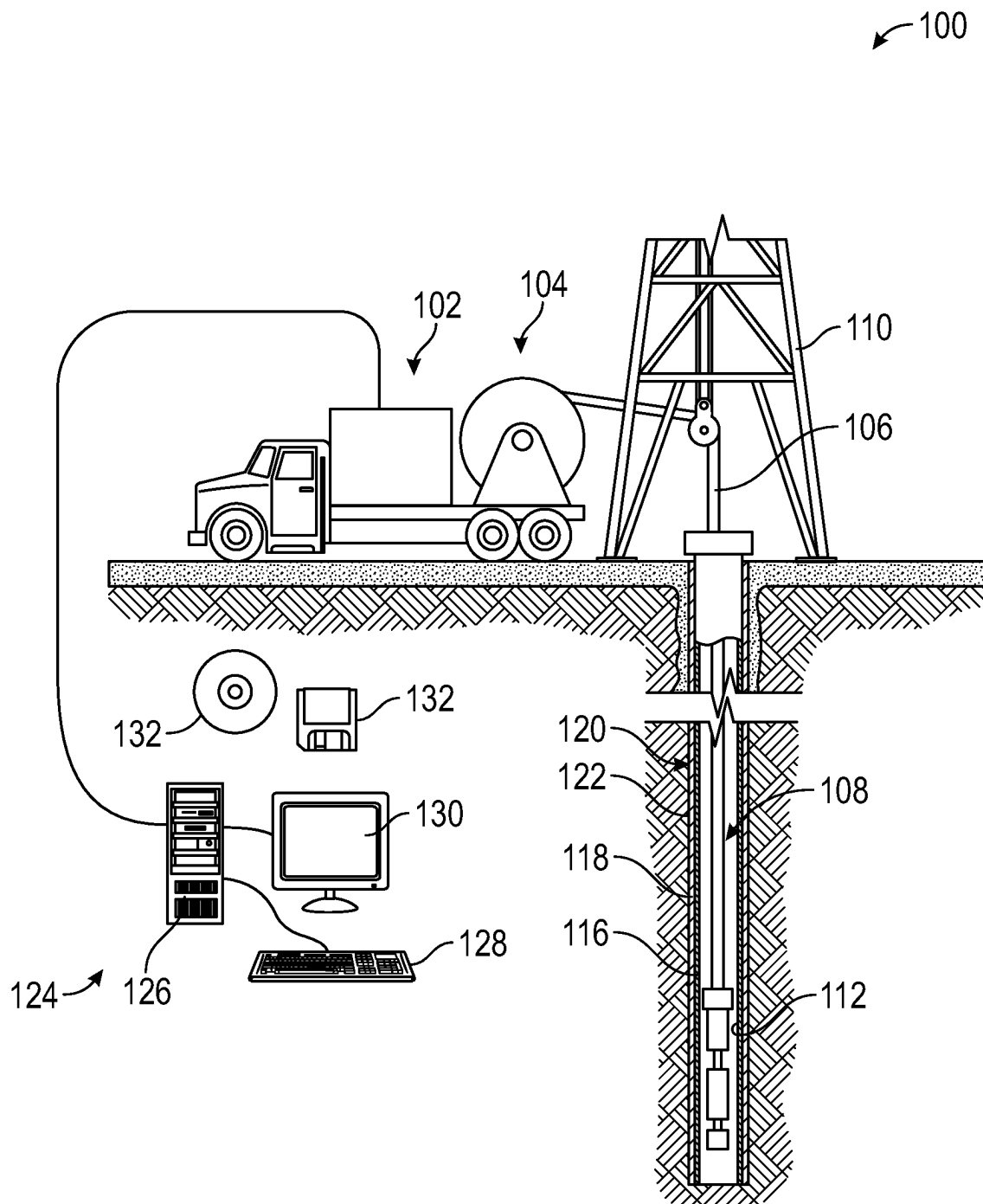


FIG. 1

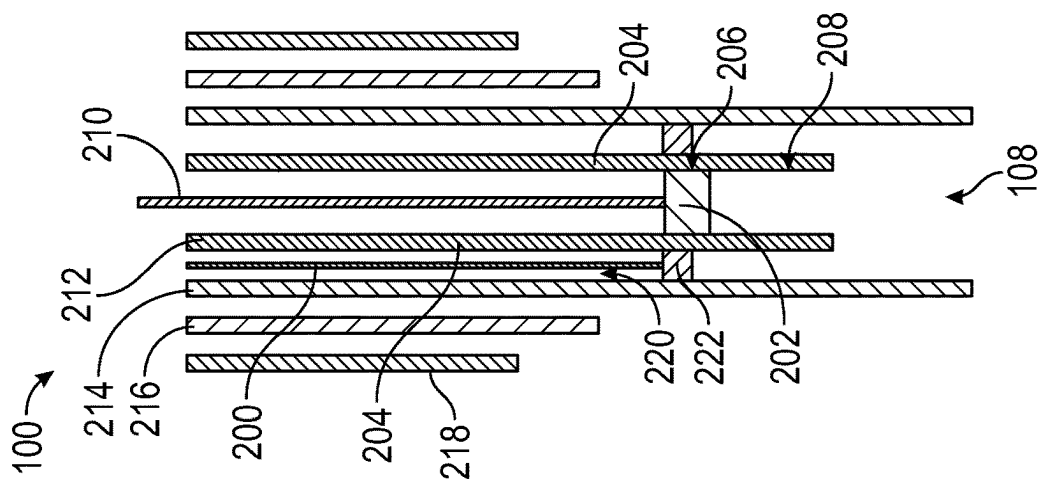


FIG. 2A

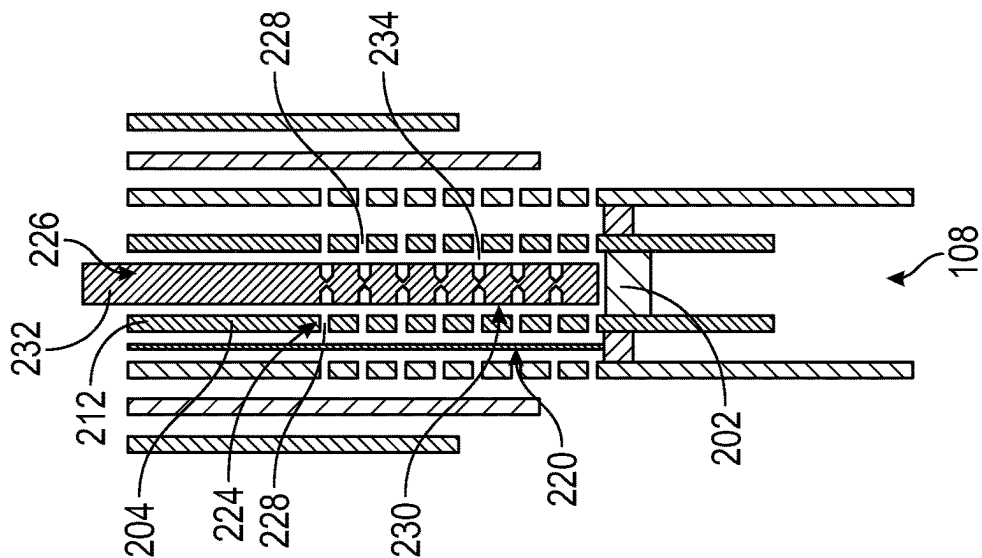


FIG. 2B

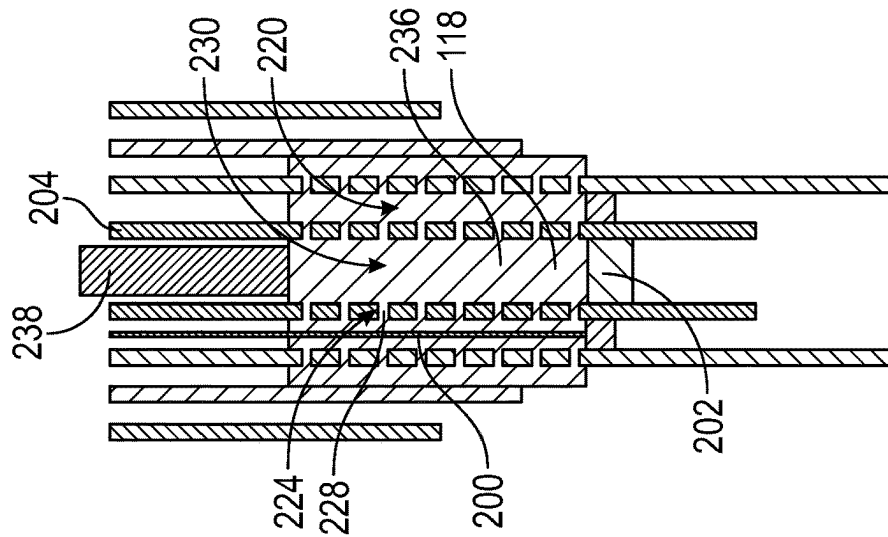


FIG. 2C

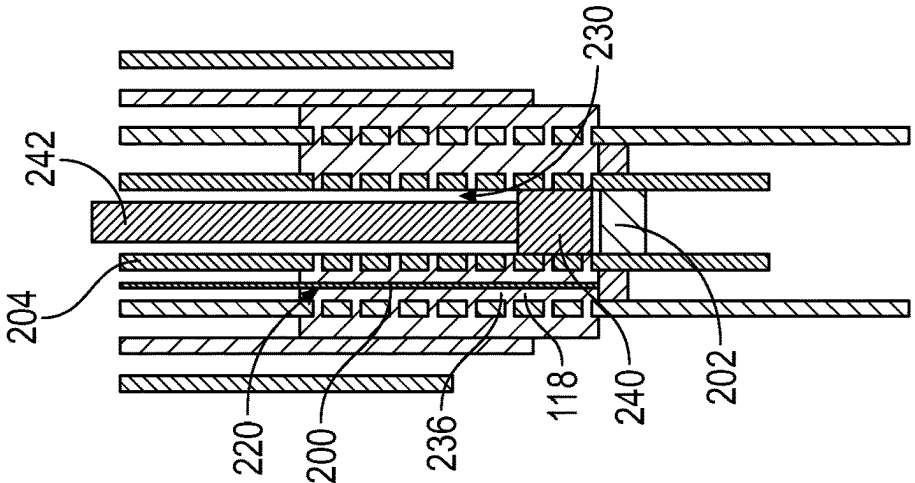


FIG. 2D

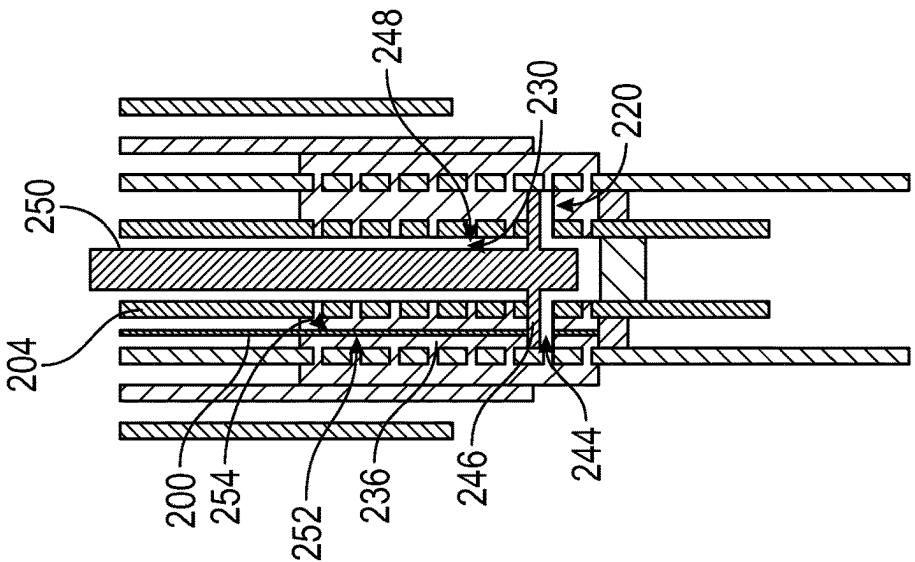


FIG. 2E

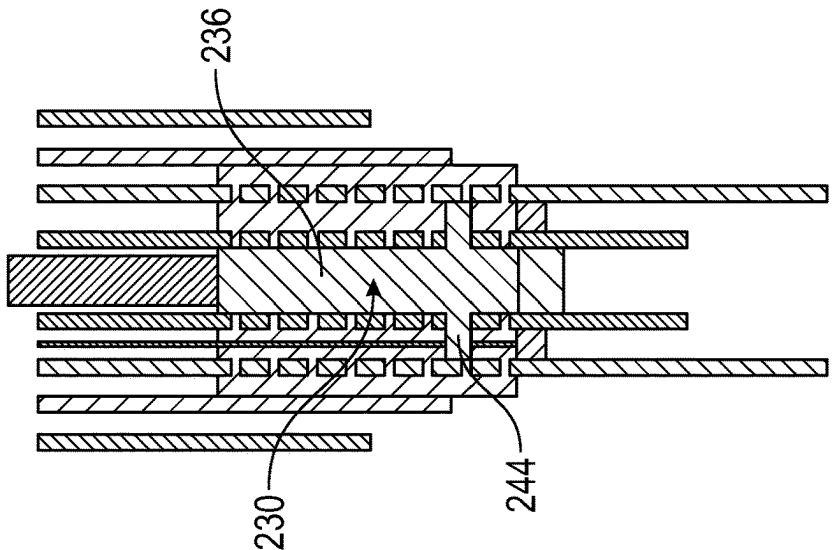


FIG. 2F

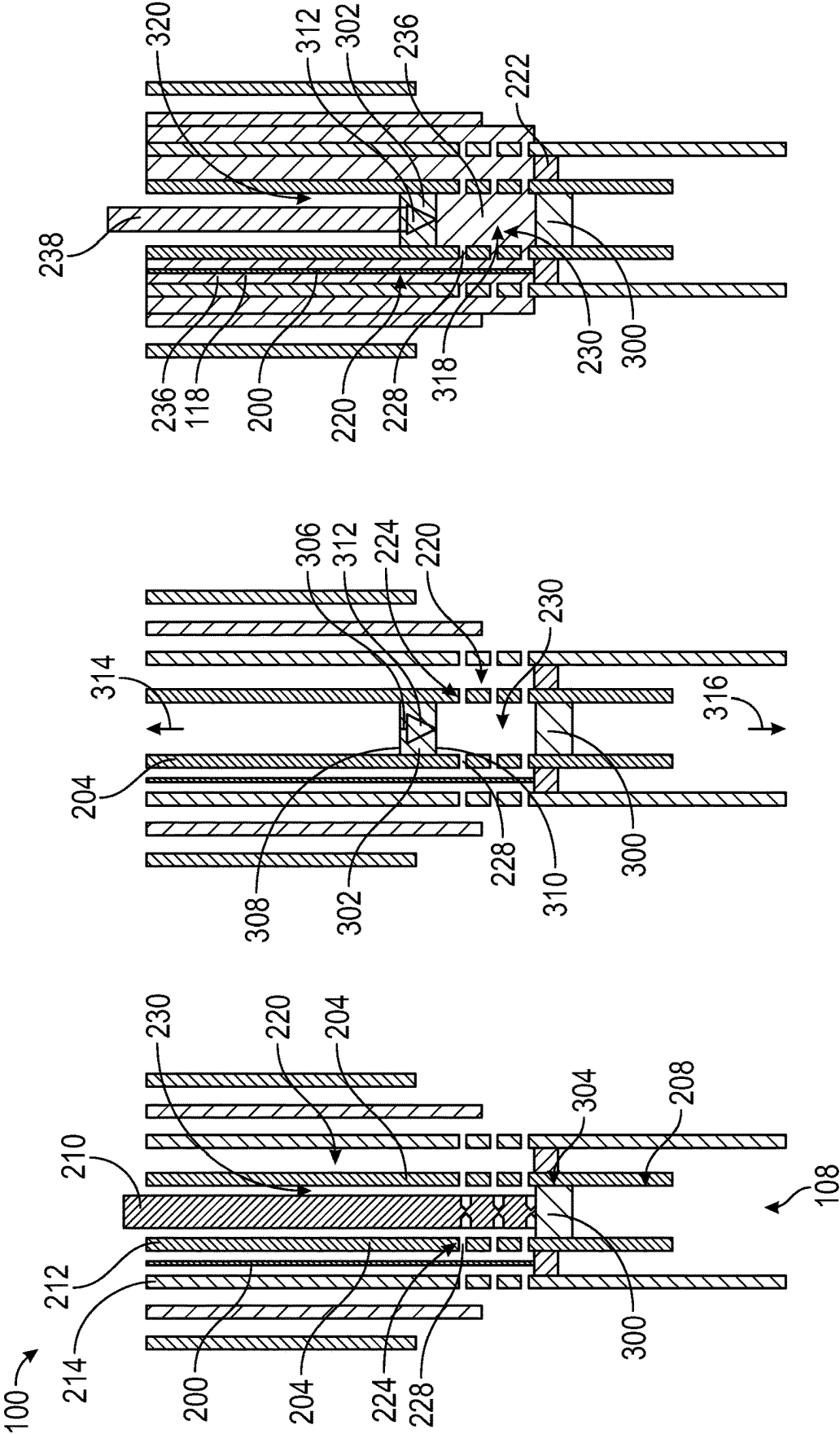


FIG. 3C

FIG. 3B

FIG. 3A

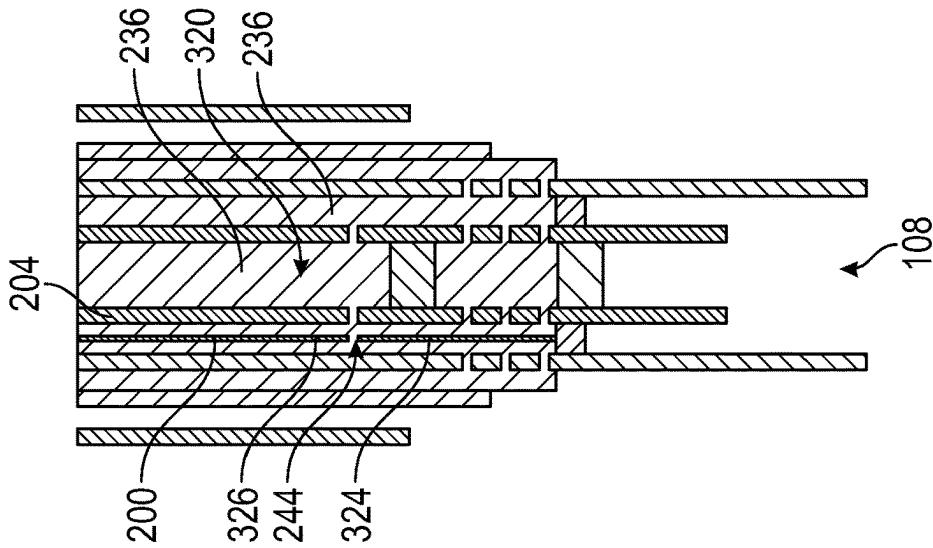


FIG. 3F

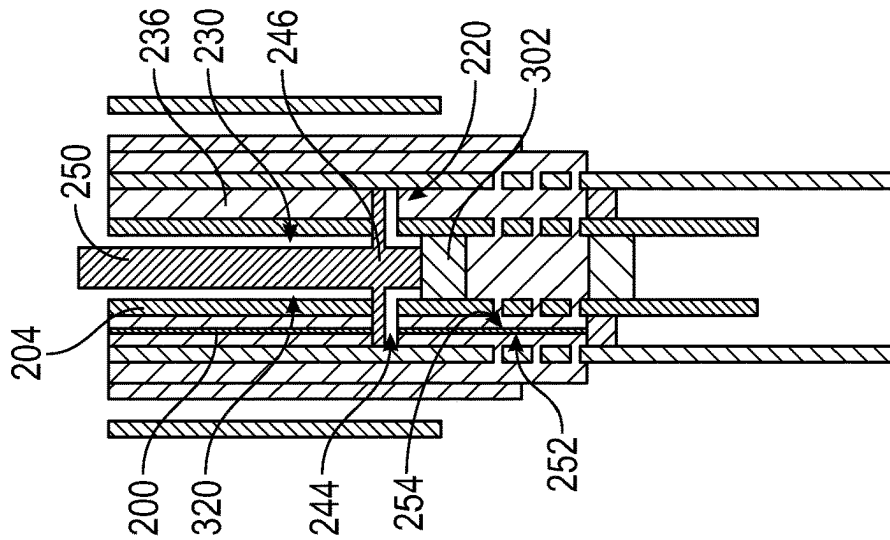


FIG. 3E

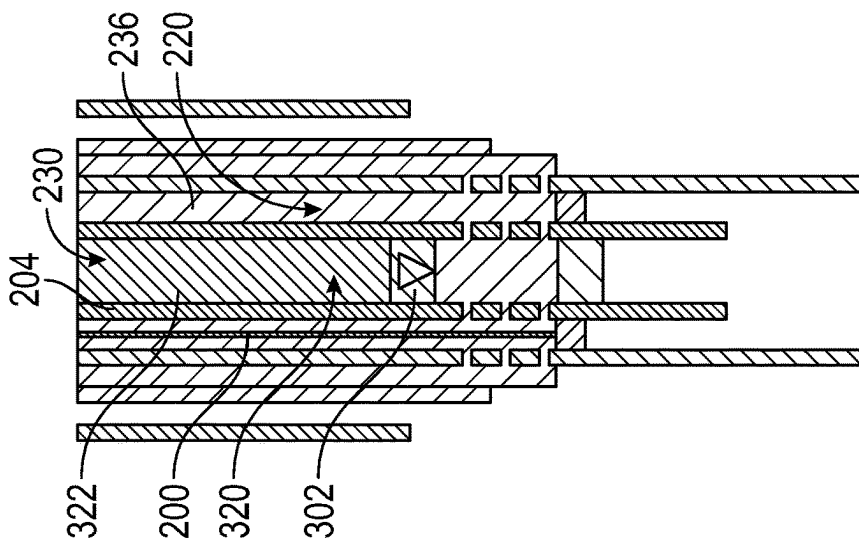


FIG. 3D

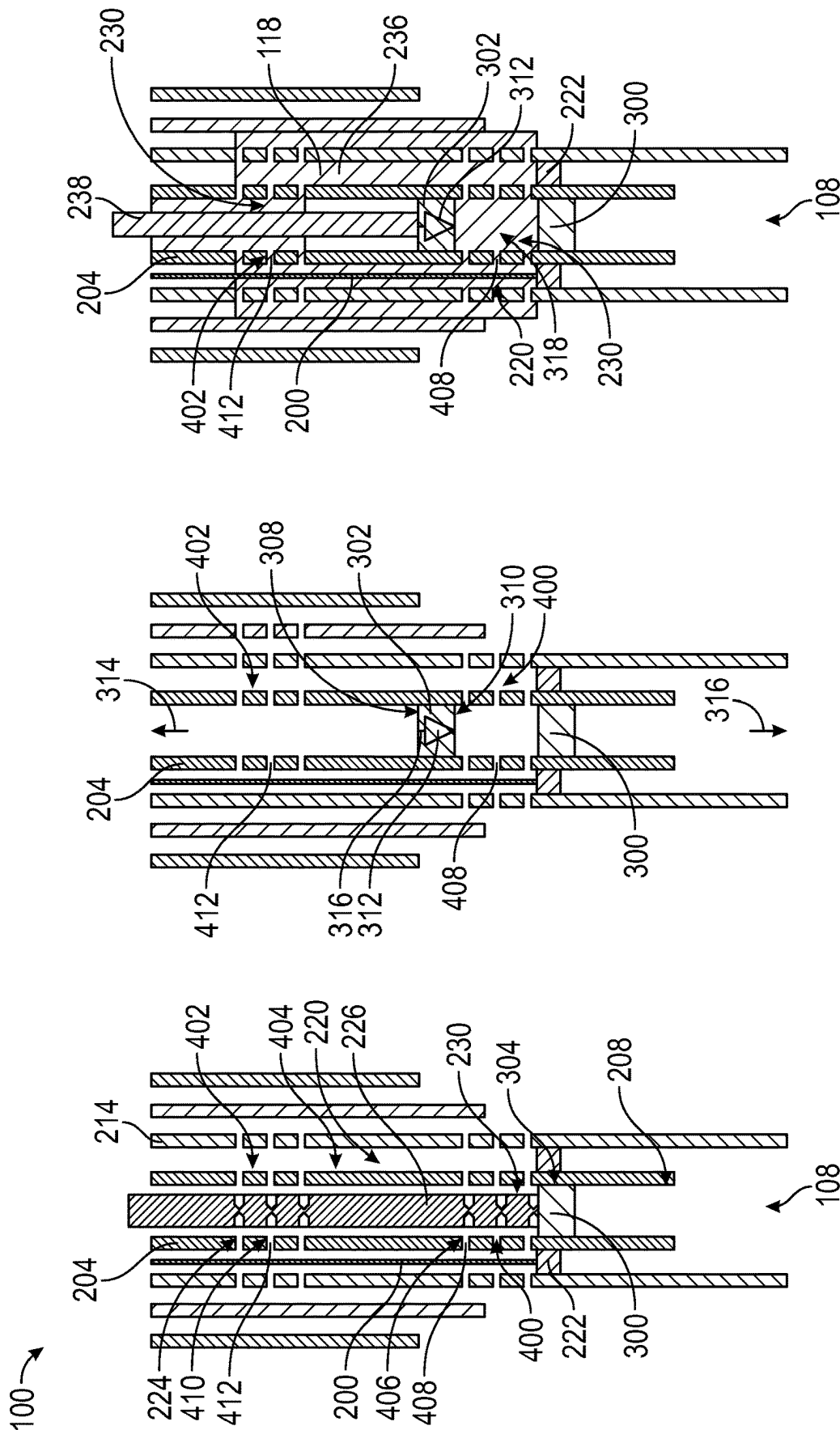


FIG. 4A

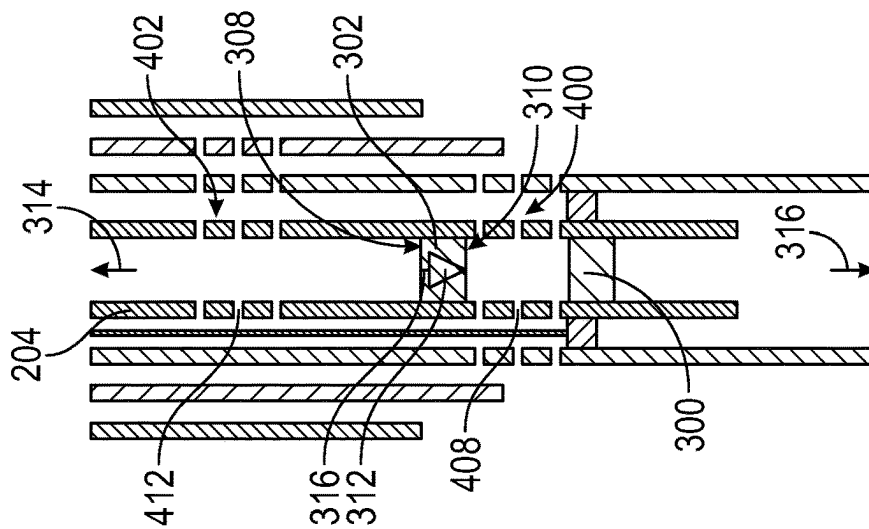


FIG. 4B

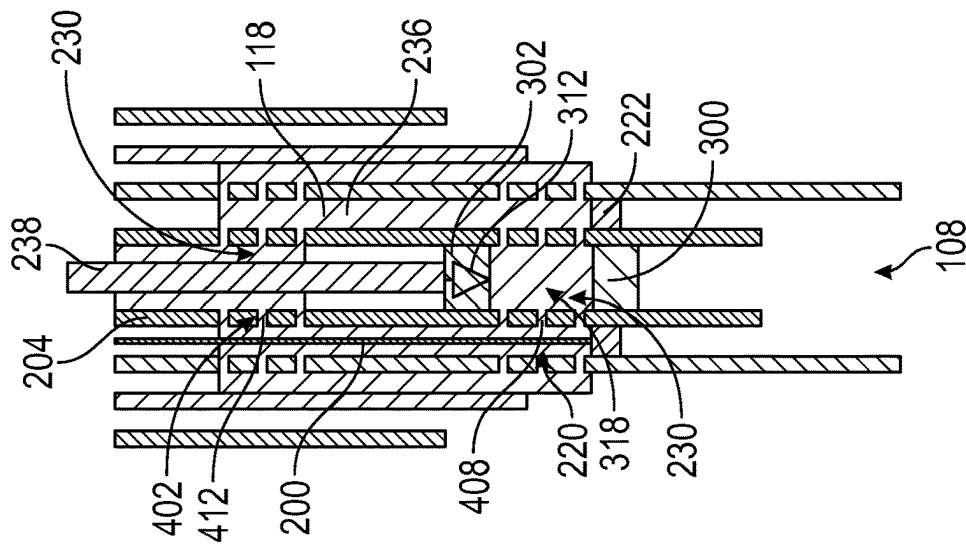


FIG. 4C

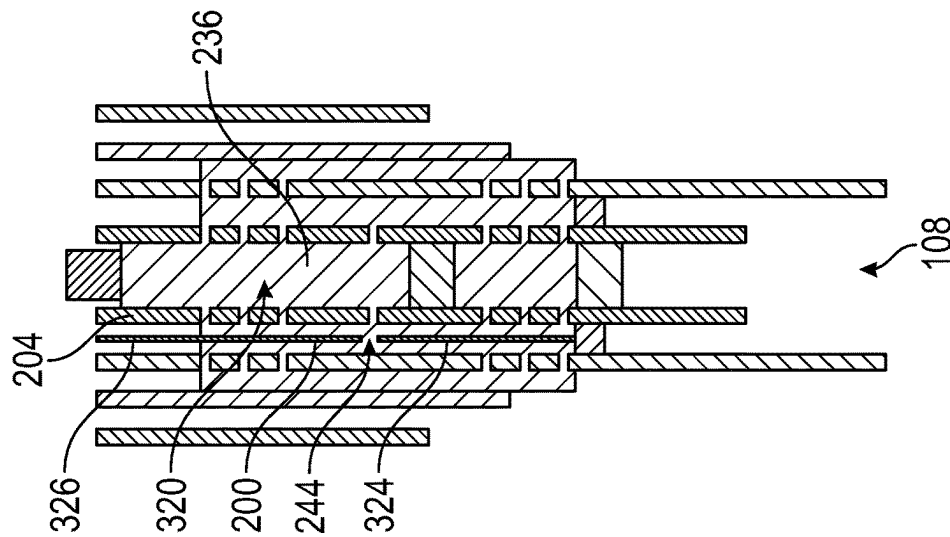


FIG. 4F

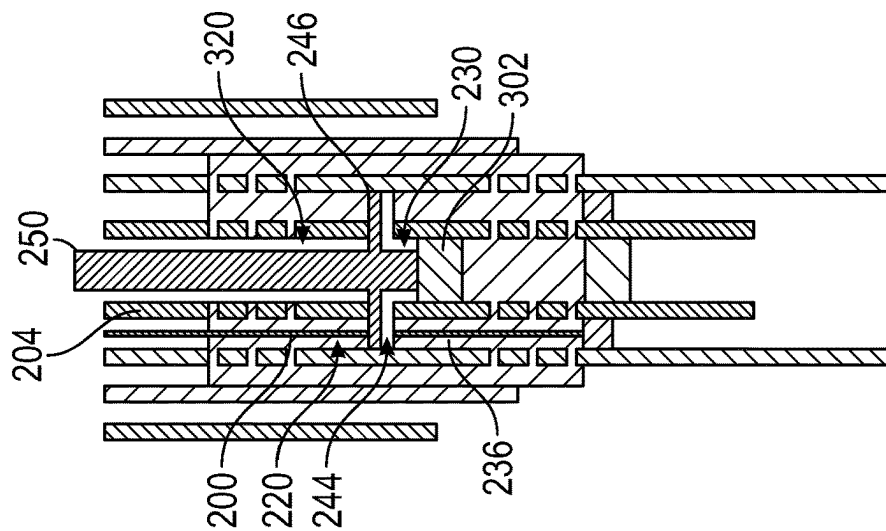


FIG. 4E

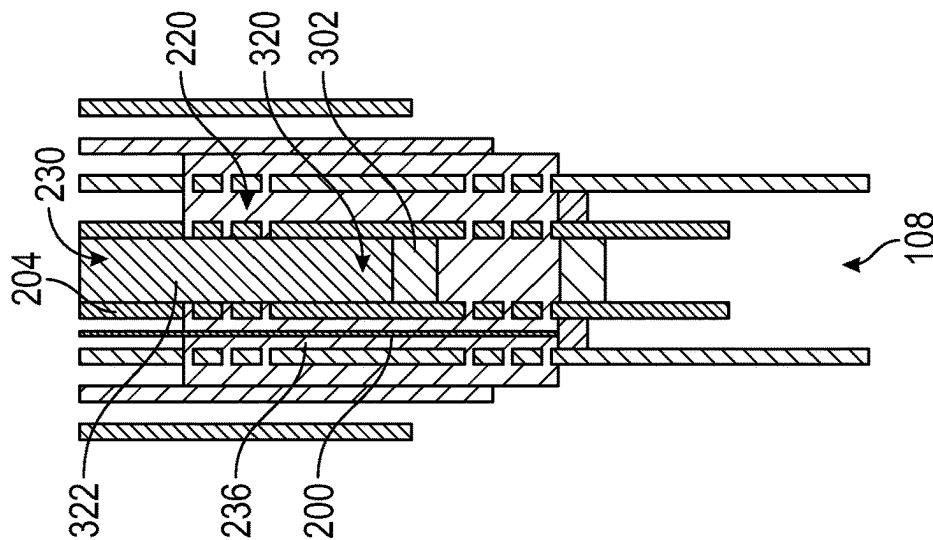


FIG. 4D

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WELL ABANDONMENT AND SEVERANCE OF CONTROL LINES

BACKGROUND

After drilling a wellbore in a subterranean formation for recovering hydrocarbons such as oil and gas lying beneath the surface, a casing string may be fed into the wellbore. Generally, the casing string protects the wellbore from failure (e.g., collapse, erosion) and provides a fluid path for hydrocarbons during production. Further, cement may be pumped into the annular space between the casing and a wellbore wall to form a seal. To access the hydrocarbons for production, a perforating gun system may be deployed into the casing string to form perforations in the casing and/or cement such that hydrocarbons may flow into the casing string via the perforation. Further, tubing may be lowered into the well as part of production to provide a reduced diameter fluid path for the hydrocarbons during production. Once production operations have concluded, plug and abandonment (P&A) operations may be conducted. P&A operations generally include severing control lines and/or flat packs to reduce sealing issues for the wellbore. Non-severed control lines may provide leak paths for hydrocarbons to travel through a cement seal formed in the wellbore as part of the P&A operations.

Generally, during P&A operations, a cutting tool is lowered into the wellbore to cut the control lines prior to pumping cement into the wellbore since the control lines are generally accessible at this stage. In particular, the cutting tool is lowered to a desired depth in the tubing where the cutting tool is deployed to cut through the tubing and the control lines at the desired depth. However, control lines may float in the annulus between the tubing and the casing. As such, the control lines may move in response to contact with the cutting tool, which may result in the cutting tool failing to successfully sever the control lines. Unfortunately, as set forth above, failing to sever the control lines may result in leak paths extending through the cement, which may compromise the seal formed by the P&A operations.

BRIEF DESCRIPTION OF THE DRAWINGS

These drawings illustrate certain aspects of some of the embodiments of the present disclosure and should not be used to limit or define the method.

FIG. 1 illustrates an elevation view of a well system, in accordance with some embodiments of the present disclosure.

FIGS. 2A-2F illustrate respective cross-sectional views of a well abandonment system configured to sever control lines in a wellbore, in accordance with some embodiments of the present disclosure.

FIGS. 3A-3F illustrate respective cross-sectional views of another well abandonment system configured to sever control lines in a wellbore, in accordance with some embodiments of the present disclosure.

FIGS. 4A-4F illustrate respective cross-sectional views of another well abandonment system configured to sever control lines in a wellbore, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

Disclosed herein are systems and methods for severing control lines and/or flat packs during plug and abandonment (P&A) operations and, more particularly, example embodi-

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ments may include injecting a sealing material such as cement into an annulus formed between a wellbore wall of the wellbore and a tubular disposed in the wellbore. The control lines and/or flat packs may be disposed in the annulus. As set forth in greater detail below, the sealing material may be configured to restrain movement of the control lines and/or flat packs such that an orbital cut from a cutting device may reliably sever the control lines and/or flat packs.

FIG. 1 illustrates an elevation view of a well system, in accordance with some embodiments of the present disclosure. It should be noted that while FIG. 1 generally depicts a land-based operations, those skilled in the art will readily recognize that the principles described herein are equally applicable to subsea operations that employ floating or sea-based platforms and rigs, without departing from the scope of the disclosure. As illustrated, a well abandonment system 100 may comprise a vehicle 102 (e.g., mounted truck coiled tubing unit) or any suitable surface system 104 for raising and lowering coiled tubing 106 in a wellbore 108. Generally, the wellbore 108 may be lined with casing 116, and cement 118 may be pumped into an outer annulus 120 between the casing 116 and a wellbore wall 122 to protect the wellbore 108 from failure (e.g., collapse, erosion) and to provide a fluid path for hydrocarbons during production. Although the illustrated embodiment shows a drilling rig 110 supporting the coiled tubing 106, the vehicle 102 may be configured to run the coiled tubing 106 without a drilling rig 110. Alternatively, the vehicle and/or the drilling rig may be configured to run any suitable conveyance (e.g., slickline, wireline, drill pipe, etc.) into the wellbore. As set forth in greater detail below, the well system may lower various downhole tools 112 (e.g., perforating guns, cutting devices, milling devices, etc.) via a corresponding conveyance.

Moreover, during plug and abandonment operations, the well abandonment system 100 may relay information between the surface and the downhole tools 112. In particular, the well abandonment system 100 may further include an information handling system 124 configured to process information gathered by the downhole tools 112. For example, sensor data recorded by downhole tools 112 may be communicated to and then processed by information handling system 124. Without limitation, the processing may be performed in real-time. Processing may alternatively occur downhole or may occur both downhole and at the surface. The sensor data recorded by downhole tools 112 may be conducted to information handling system 124 via coiled tubing 106 or any suitable transmission medium. Information handling system 124 may process the signals, and the information contained therein may be displayed for an operator to observe and stored for future processing and reference. Information handling system 124 may also contain an apparatus for supplying control signals to the downhole tools 112.

Information handling system 124 may include any instrumentality or aggregate of instrumentalities operable to compute, estimate, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system 124 may be a processing unit 126, a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. Information handling system 124 may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM,

and/or other types of nonvolatile memory. Additional components of the information handling system **124** may include one or more disk drives, one or more network ports for communication with external devices as well as various input and output (I/O) devices, such as an input device **128** (e.g., keyboard, mouse, etc.) and a display **130**. Information handling system **124** may also include one or more buses operable to transmit communications between the various hardware components.

Alternatively, systems and methods of the present disclosure may be implemented, at least in part, with non-transitory computer-readable media **132**. Non-transitory computer-readable media **132** may include any instrumentality or aggregation of instrumentalities that may retain data and/or instructions for a period of time. Non-transitory computer-readable media **132** may include, for example, storage media such as a direct access storage device (e.g., a hard disk drive or floppy disk drive), a sequential access storage device (e.g., a tape disk drive), compact disk, CD-ROM, DVD, RAM, ROM, electrically erasable programmable read-only memory (EEPROM), and/or flash memory; as well as communications media such wires, optical fibers, microwaves, radio waves, and other electromagnetic and/or optical carriers; and/or any combination of the foregoing.

FIGS. 2A-2F illustrate respective cross-sectional views of the well abandonment system **100** configured to sever control lines **200** in the wellbore **108**, in accordance with some embodiments of the present disclosure. In particular, FIG. 2A discloses the wellbore **108** having a plug **202** disposed in an inner tubular **204** (e.g., production tubing) to form a fluid barrier. That is, the plug **202** may be run-in-hole and set at a target location in the inner tubular **204** to block fluid from traveling through the inner tubular **204**. The plug **202** may include any suitable material. In the illustrated embodiment, the plug comprises a cylindrical shape such that a radially outer surface **206** of the plug **202** may seal against a radially inner surface **208** of the inner tubular **204** to restrain fluid. However, the plug may include any suitable shape for blocking fluid. Further, the plug **202** may be expandable. That is, the plug **202** may be run-in-hole to the target location and then expand at the target location to seal against the inner tubular **204**.

The plug **202** may be run-in-hole via a conveyance (e.g., plug conveyance **210**) to the target location (e.g., a predetermined sealing position in the wellbore **108**). Sealing positions (e.g., position for sealing the wellbore during plug and abandonment operations) may be predetermined based at least in part on local regulations, common practice, or other suitable guidance. For example, local regulations may require that a wellbore be sealed between production zones and usable water strata. As such, prior to plug and abandonment operations, a sealing position between a production zone and usable water strata may be predetermined. Moreover, in the illustrated embodiment, the plug conveyance **210** includes a slickline configured to lower the plug **202** into the wellbore **108**. However, the plug conveyance **210** may include any suitable conveyance, such as a wireline, coiled tubing, drill pipe, or some combination thereof, for running the plug **202** into the wellbore **108**.

Moreover, the wellbore **108** may be formed with various tubulars (e.g., production tubing **212**, production casing **214**, intermediate casing, surface casing **216**, conductor casing **218**, etc.). In the illustrated embodiment, the conductor casing **218** is the radially outermost tubular disposed in the wellbore **108**. The surface casing **216** may be disposed within the conductor casing **218** (e.g., disposed radially

inward from the conductor casing **218**), and the production casing **214** may be disposed within the surface casing **216**. Further, the inner tubular **204** (e.g., production tubing **212**) is disposed within the production casing **214**. However, the wellbore **108** may include any suitable structure. For example, the wellbore **108** may include at least one intermediate casing disposed between the surface casing **216** and the production casing **214**. During installation of the tubulars, cement **118** may be disposed at least between the conductor casing **218** and the wellbore wall **122**, between the surface casing **216** and the conductor casing **218**, and between the production casing **214** and the surface casing **216**. However, an annulus **220** formed between the inner tubular **204** and the production casing **214** may be left open such that fluid may flow through the annulus **220**. In the illustrated embodiment, the annulus **220** may be sealed via a packer assembly **222** or any suitable sealing feature to restrain fluid flow from a bottom of the wellbore **108** to the surface. However, some wellbores **108** may not include the packer assembly **222**. Further, as illustrated, the plug **202** may be run-in-hole to a location disposed proximate the packer assembly **222**. The packer assembly **222** and the plug **202**, in combination, may provide a fluid barrier between the surface and a bottom of the wellbore **108** and/or a production zone.

FIG. 2B discloses the inner tubular **204** having a plurality of perforations **224**. With the plug **202** set in the inner tubular **204** (e.g., the production tubing), a perforating device **226** may be lowered into the wellbore **108** to perforate the inner tubular **204** at a position above the plug **202**. Perforating the inner tubular **204** above the plug **202** may open at least one fluid path **228** between a central bore **230** of the inner tubular **204** and the annulus **220**. As set forth above, the annulus **220** may be formed between the inner tubular **204** and the production casing **214** or wellbore wall **122**. As set forth in greater detail below, the at least one fluid path **228** may fluidly connect the central bore **230** to the annulus **220** at a location proximate the plug **202** such that sealing material **236** (shown in FIG. 2C) may be injected into the annulus **220** from the central bore **230**.

Moreover, the perforating device **226** may include a perforating gun **232** having shaped charges **234** configured to detonate in a substantially radially outward direction to perforate the inner tubular **204** above the plug **202** disposed in the wellbore **108**. In the illustrated embodiment, both the inner tubular **204** (e.g., production tubing **212**) and the production casing **214** have been perforated by the perforating device **226**. For example, a blast from the shaped charges **234** may perforate the inner tubular **204**, the production casing **214**, and continue through the wellbore wall **122** into a downhole formation. Alternatively, the perforating device **226** may be configured to only perforate the inner tubular **204**. For example, the perforating device **226** may comprise a cutting device **246** (e.g., blade, reamer, hydro jet, plasma torch, etc.) shown in FIG. 2E. The cutting device **246** is configured to make at least one orbital cut to perforate the inner tubular **204** (e.g., production tubing) above the plug **202** disposed in the wellbore **108**. However, the cutting device **246** may be configured to also perforate the production casing **214** and/or the wellbore wall **122**.

Moreover, as illustrated, the perforating device **226** may be configured to make a plurality of perforations to open a plurality of fluid paths **228** between the central bore **230** and the annulus **220**. For example, the perforating device **226** may comprise a perforating gun **232** having a dozen shaped charges **234**. In response to detonation of the shaped charges **234**, the inner tubular **204** may be perforated with a dozen

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corresponding perforations 224 that each form a respective fluid path 228 between the central bore 230 and the annulus 220. As set forth in greater detail below, having a plurality of fluid paths 228 may help increase a flow rate of fluid (e.g., the sealing material 236 shown in FIG. 2C) between the central bore 230 and the annulus 220. The perforating device 226 may be configured to form perforations 224 in the inner tubular 204 with any suitable spacing, sizing, and number, to provide a desired flow area between the central bore 230 and the annulus 220.

FIG. 2C discloses the sealing material 236 disposed in the annulus 220 and the central bore 230 of the inner tubular 204. With the inner tubular 204 perforated above the plug 202, the sealing material 236 may be injected into the wellbore 108. In particular, an injection conveyance 238 (e.g., coiled tubing, drill pipe, etc.) may be lowered into the inner tubular 204 to a position proximate the plug 202 and/or the at least one perforation 224 in the inner tubular 204. The sealing material 236 may be delivered in an at least partially liquid state to the central bore 230. As such, the sealing material 236 may flow downhole toward the plug 202 and flow radially outward into the annulus 220 via the at least one fluid path 228. As such, the sealing material 236 may be injected into the annulus 220 formed between the inner tubular 204 (e.g., production tubing) and the production casing 214 and/or the wellbore wall 122. The sealing material 236 is configured to surround at least a portion of the control line 200 and/or flat pack disposed in the annulus 220.

Further, the sealing material 236 may include cement, plastic, resin, or some combination thereof. As set forth above, the sealing material 236 is injected into the annulus 220 in at least a partially liquid or fluid state. However, the sealing material 236 is configured to solidify after being injected into the annulus 220. For example, the sealing material 236 may include cement 118 that is mixed with water prior to being injected into the wellbore 108 such that the cement 118 is in a liquid or fluid state. In the fluid state, the cement 118 may flow through the injection conveyance 238 down to the central bore 230 and out through the at least one fluid path 228 into the annulus 220. Due to the chemical reaction (e.g., hydration) between the water and the cement 118, the cement 118 may solidify or harden over time. Once hardened, the cement 118 disposed in the annulus 220 may restrain movement of the control line 200. Specifically, the hardened cement 118 may restrain lateral movement of the control line 200.

FIG. 2D discloses the sealing material 236 being milled out from the central bore 230 of the inner tubular 204. With the sealing material 236 disposed and hardened in the annulus 220, the sealing material 236 may be configured to restrain movement of the control lines 200 and/or flat packs such that making an orbital cut 244 (shown in FIG. 2E) with the cutting device 246 may reliably sever the control lines 200 and/or flat packs. However, to make the orbital cut, the cutting device 246 may need to be lowered into the inner tubular 204 to a location in the central bore 230 that is axially aligned with a portion of the annulus 220 that is filled with the sealing material 236. Unfortunately, injecting the sealing material 236 into the annulus 220 may also dispose the sealing material 236 in the central bore, and the sealing material 236 disposed in the central bore 230 may be at a same height or higher than the sealing material 236 in the annulus 220.

Thus, to lower the cutting device 246 into the inner tubular 204 to a location that is axially aligned with a portion of the annulus 220 that is filled with the sealing material 236, at least a portion the sealing material 236 in the central bore

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230 may need to be removed. For example, a milling device 240 may be lowered into the central bore 230 to mill out at least a portion of the hardened cement 118 disposed in the central bore 230 above the plug 202. As illustrated, the milling device 240 may mill out the cement 118 down to a location of the plug 202. Alternatively, the milling device 240 may mill out only a portion of the cement 118 in the central bore 230. However, the milling device 240 may at least mill out a sufficient portion of the cement 118 in the central bore 230 such that the cutting device 246 may be lowered to a position in the central bore 230 that is axially aligned with a portion of the annulus 220 having the hardened cement 118 surrounding the control lines 200 and/or flat pack.

Moreover, the milling device 240 may include a drill bit (e.g., a fixed cutter drill bit, a roller cone bit, a hybrid drill bit, etc.), a reamer, or any other suitable milling device. The milling device 240 may be run-in-hole via a milling conveyance 242 (e.g., drill pipe, coiled tubing, etc.) to the hardened cement 118 in the central bore 230. With the milling device 240 in position, a mud motor (not shown) may be configured to drive rotation of the milling device 240 to engage and mill out the cement 118. Once the cement 118 in the central bore is sufficiently milled out, the milling conveyance 242 may be configured to pull the milling device 240 out-of-hole.

FIG. 2E discloses the control line 200 and/or flat pack severed via making an orbital cut 244. As illustrated, the cutting device 246 may be run-in-hole to a target location in the central bore 230 for making the at least one orbital cut 244 (e.g., a location that is axially aligned with a portion of the annulus 220 that is filled with the hardened sealing material 236). For example, the cutting device 246 may be run-in-hole to a milled out portion 248 of the central bore 230 of the inner tubular 204 to make the at least one orbital cut 244. The cutting device 246 may be run-in-hole via a cutting device conveyance 250 (e.g., a slickline, a wireline, coiled tubing, drill pipe, etc.). Further, the cutting device 246 may include any suitable cutting device. For example, the cutting device 246 may include an abrasive jetting device, a blade cutter, a reamer, a plasma torch, or some combination thereof.

Moreover, at the target location, the cutting device 246 may be configured to cut through the inner tubular 204 (e.g., production tubing) and the sealing material 236 in the annulus 220 and sever the control line 200 surrounded by the sealing material 236. Indeed, the hardened sealing material 236 is configured to restrain at least lateral movement of the control line 200. As set forth above, the cutting device 246 may fail to cut the control line 200 if the control line 200 is permitted to move in response to contact with the cutting device 246. However, in the illustrated embodiment, as the cutting device 246 cuts through the sealing material 236 and engages the control line 200 and/or flat pack, the remaining sealing material 236 may prevent the control line 200 and/or flat pack from moving laterally in response to contact with the cutting device 246. That is, the cutting device 246 may engage a radially inner side 252 of the control line 200 and a radially outer side 254 of the control line 200 may be in contact with the remaining hardened sealing material 236. As the cutting device 246 engages the radially inner side 252, the remaining hardened sealing material 236 may prevent the control line 200 from moving radially outward such that the cutting device 246 may engage and sever the control line 200. Similarly, should the cutting device 246 engage another side of the control line 200, an opposite side of the control line 200 may interface with the remaining

hardened sealing material **236** to prevent the control line **200** from moving with respect to the cutting device **246**.

FIG. 2F discloses the sealing material **236** (e.g., cement) disposed in the orbital cut **244** and the central bore **230**. After making the orbital cut **244**, additional sealing material **236** may be injected into the central bore **230** of the inner tubular **204** (e.g., production tubing) to fill the volume removed by making the orbital cut **244** and at least a portion of the central bore **230**. As set forth above, the control lines **200** may provide leak paths for hydrocarbons to travel through a seal formed in the wellbore as part of the P&A operations. Accordingly, filling the orbital cut **244** with the additional sealing material **236** may fluidly isolate a lower portion **324** of the severed control line **200** from an upper portion **326** of the severed control line **200**, which may block a leak path for hydrocarbons.

FIGS. 3A-3F illustrate respective cross-sectional views of another well abandonment system **100** configured to sever the control lines **200** in the wellbore **108**, in accordance with some embodiments of the present disclosure. FIG. 3A discloses the wellbore **108** having a primary plug **300** disposed in an inner tubular **204** to form a fluid barrier, as well as the inner tubular **204** having a plurality of perforations **224**. The primary plug **300** may be run-in-hole and set at a target location in the inner tubular **204** (e.g., the production tubing) to block fluid from traveling through the inner tubular **204** at the primary plug **300**. The primary plug **300** may include any suitable material. In the illustrated embodiment, the primary plug **300** comprises a cylindrical shape such that a radially outer surface **304** of the primary plug **300** may seal against the radially inner surface **208** of the inner tubular **204** to restrain fluid. However, the primary plug **300** may include any suitable shape for blocking fluid. Further, the primary plug **300** may be expandable. That is, the primary plug **300** may be run-in-hole to the target location and then expand at the target location to seal against the inner tubular **204**.

Further, the annulus **220** may be formed between the inner tubular **204** and the production casing **214** and/or the wellbore wall **122** may be left open (e.g., not filled with cement) such that fluid may flow through the annulus **220**. In the illustrated embodiment, the annulus **220** may be sealed via the packer assembly **222** or any suitable sealing feature to restrain fluid flow from the bottom of the wellbore **108** to the surface. Further, as illustrated, the primary plug **300** may be run-in-hole to a location disposed proximate the packer assembly **222**. The packer assembly **222** and the primary plug **300**, in combination, may provide a fluid barrier between the surface and the bottom of the wellbore **108** and/or a production zone.

Moreover, with the primary plug **300** set in the inner tubular **204**, the perforating device **226** (e.g., perforating gun, blade, reamer, hydro jet, plasma torch, etc.) may be lowered into the wellbore **108** to perforate the inner tubular **204** above the primary plug **300**. Perforating the inner tubular **204** above the primary plug **300** may open the at least one fluid path **228** between the central bore **230** of the inner tubular **204** and the annulus **220**. As set forth above, the annulus **220** may be formed between the inner tubular **204** and the production casing **214** and/or the wellbore wall **122**. The at least one fluid path **228** may fluidly connect the central bore **230** to the annulus **220** at a location proximate the primary plug **300** such that the sealing material **236** may be injected into the annulus **220** from the central bore **230**. In the illustrated embodiment, the inner tubular **204** has a

plurality of perforations **224** such that a plurality of fluid paths **228** extend between the central bore **230** and the annulus **220**.

FIG. 3B discloses a secondary plug **302** disposed in the inner tubular **204**. As illustrated, the secondary plug **302** may be run-in-hole and set in a position uphole from both the primary plug **300** and the plurality of perforations **224** (e.g., the plurality of fluid paths **228** extending between the central bore **230** and the annulus **220**). Moreover, the secondary plug **302** may comprise a through bore **306** extending from an uphole end **308** to a downhole end **310** of the secondary plug **302** such that fluid may pass through the secondary plug **302**. However, the secondary plug **302** may also include a check-valve **312** configured to restrain fluid flow in the uphole direction **314**. The check-valve **312** may include any suitable type of check valve for restraining flow of the sealing material **236** in the uphole direction **314** and permitting fluid flow of the sealing material **236** in a downhole direction **316** through the check-valve **312**. For example, the check-valve **312** may include a flapper check valve having a flapper (not shown) configured to swing open in response to fluid (e.g., the sealing material **236**) flowing in the downhole direction **316** and close in response to the fluid flowing in the uphole direction **314**. Alternatively, a closing sleeve (not shown) may be disposed in the secondary plug **302** to restrain fluid flow in the uphole direction **314** while permitting fluid flow in the downhole direction **316**. That is, the closing sleeve may open in response to the injection conveyance **238** stinging into the closing sleeve. With the closing sleeve connected to the injection conveyance **238**, downhole flow of the sealing material into the closing sleeve from the injection conveyance **238** may block fluid flowing in the uphole direction **314**. Further, in response to the injection conveyance **238** disconnecting (e.g., sting out) the closing sleeve is configured to close to block fluid flow in the uphole direction **314**. However, any suitable device may be used to restrain fluid flow in the uphole direction **314** while permitting fluid flow in the downhole direction **316** through the secondary plug **302**.

FIG. 3C discloses the sealing material **236** disposed in the annulus **220** and a portion of the central bore **230** disposed between the primary plug **300** and the secondary plug **302** (e.g., a lower central bore portion **318**). As illustrated, the sealing material **236** (e.g., cement) may be pumped through the check-valve **312** into the lower central bore portion **318**. Further, as illustrated, the sealing material **236** may be injected through the secondary plug **302** via the injection conveyance **238** (e.g., drill pipe, coiled tubing, or some combination thereof) to help minimize an amount of the sealing material **236** injected into the portion of the central bore **230** disposed uphole from the secondary plug **302** (e.g., an upper central bore portion **320**). As set forth in greater detail below, the cutting device **246** (shown in FIG. 3E) may make the orbital cut **244** (shown in FIG. 3E) above the secondary plug **302**, so maintaining the upper central bore portion **320** clear of sealing material **236** may allow the cutting device **246** to be lowered into position for making the orbital cut **244** without having to mill out the upper central bore portion **320** with the milling device **240** (shown in FIG. 2D). In some embodiments, the sealing material **236** may be pushed downhole through the secondary plug **302** via a dart (not shown) driven downhole with fluid pressure from a fluid (e.g., drilling fluid, cleaning fluid, etc.) injected into the wellbore **108**. The fluid may provide fluid pressure to drive the dart while also cleaning the upper central bore portion **320** as the dart moves toward the secondary plug **302**.

Moreover, the sealing material **236** may be pumped into the lower central bore portion **318** in a partially liquid state. Accordingly, as the lower central bore portion **318** is filled with the sealing material **236**, fluid pressure from the sealing material **236** entering through the check-valve **312** may drive the sealing material **236** through the at least one fluid path **228** and into the annulus **220**. The sealing material **236** may continue to be pumped through the check-valve **312** to drive the sealing material **236** to fill a portion of the annulus **220** proximate the packer assembly **222** and continue to flow uphole in the annulus **220** to fill the annulus **220** up to a position uphole from the secondary plug **302**. That is, sealing material **236** may be pumped into the annulus **220** until the annulus **220** is at least sufficiently filled such that the sealing material **236** surrounds at least a portion of a control line **200** disposed uphole from the secondary plug **302**.

Further, as set forth above, the sealing material **236** may include cement, plastic, resin, or some combination thereof. As set forth above, the sealing material **236** may be pumped into the lower central bore portion **318** and the annulus **220** in a partially liquid state. However, the sealing material **236** is configured to solidify after being injected into the annulus **220**. For example, the sealing material **236** may include cement **118** that is mixed with water prior to being injected into the wellbore **108** such that the cement **118** is in a liquid or fluid state. However, due to the chemical reaction (e.g., hydration) between the water and the cement **118**, the cement **118** may solidify or harden over time. Once hardened, the cement **118** disposed in the annulus **220** may restrain movement of the control line **200**. Specifically, the hardened cement **118** may restrain lateral movement of at least a portion of the control line **200** disposed uphole from the secondary plug **302**.

FIG. 3D discloses the injection conveyance **238** (shown in FIG. 3E) removed from the central bore **230** of the inner tubular **204**. Once the annulus **220** is at least sufficiently filled with the sealing material **236** such that the sealing material **236** surrounds at least a portion of a control line **200** disposed uphole from the secondary plug **302**, the injection conveyance **238** may be disconnected from the secondary plug **302** and pulled out-of-hole. Further high pressure fluid (e.g., a cleaning fluid **322**) may be run into the inner tubular **204** to remove the sealing material **236** disposed uphole from the secondary plug **302**. As set forth above, maintaining the upper central bore portion **320** clear of sealing material **236** may allow the cutting device **246** (shown in FIG. 3E) to be lowered into position for making the orbital cut **244** without having to mill out the upper central bore portion **320** with the milling device **240** (shown in FIG. 2D). Running the high pressure fluid into the central bore **230** of the inner tubular **204** may remove at least a portion of the sealing material **236** that may have entered the upper central bore portion **320** during connection and/or disconnection of the injection conveyance **238**, as well as any sealing material **236** that may have entered the upper central bore portion **320** during injection of the sealing material **236** through the secondary plug **302**.

FIG. 3E discloses the control line **200** and/or flat pack severed via making the orbital cut **244** at a position uphole from the secondary plug **302**. As illustrated, the cutting device **246** may be run-in-hole to a target location in the central bore **230** for making the at least one orbital cut **244** (e.g., a location that is axially aligned with a portion of the annulus **220** that is filled with the hardened sealing material **236**). As set forth above, the sealing material **236** is pumped into the annulus **220** to a position that is uphole from the secondary

plug **302**. Further, as set forth above, the upper central bore portion **320** (e.g., the portion of the central bore **230** disposed above the secondary plug **302**) may be substantially clear of sealing material **236**. Accordingly, with the injection conveyance **238** pulled out-of-hole, the cutting device **246** may run-in-hole to the target location above the secondary plug **302**. Indeed, the cutting device **246** may be run-in-hole to the target location above the secondary plug **302** without having to mill out the upper central bore portion **320** with the milling device **240**. The cutting device **246** may be run-in-hole via a cutting device conveyance **250** (e.g., a slickline, a wireline, coiled tubing, drill pipe, etc.). Further, the cutting device **246** may include any suitable cutting device. For example, the cutting device **246** may include an abrasive jetting device, a blade cutter, a reamer, a plasma torch, or some combination thereof.

Moreover, at the target location above the secondary plug **302**, the cutting device **246** may be configured to cut through the inner tubular **204** (e.g., production tubing) and the sealing material **236** in the annulus **220** and sever the control line **200** surrounded by the sealing material **236**. Indeed, the hardened sealing material **236** is configured to restrain at least lateral movement of the control line **200**. As set forth above, the cutting device **246** may fail to cut the control line **200** if the control line **200** is permitted to move in response to contact with the cutting device **246**. However, in the illustrated embodiment, as the cutting device **246** cuts through the sealing material **236** and engages the control line **200** and/or flat pack, the remaining sealing material **236** may prevent the control line **200** and/or flat pack from moving laterally in response to contact with the cutting device **246**. That is, the cutting device **246** may engage a radially inner side **252** of the control line **200** and the radially outer side **254** of the control line **200** may be in contact with the remaining hardened sealing material **236**. As the cutting device **246** engages the radially inner side **252**, the remaining hardened sealing material **236** may prevent the control line **200** from moving radially outward such that the cutting device **246** may engage and sever the control line **200**. Further, as set forth above, should the cutting device **246** engage another side of the control line **200**, an opposite side of the control line **200** may interface with the remaining hardened sealing material **236** to prevent the control line **200** from moving with respect to the cutting device **246**.

FIG. 3F discloses the sealing material **236** disposed in the orbital cut **244** and the upper central bore portion **320**. After making the orbital cut **244**, additional sealing material **236** may be injected into the upper central bore portion **320** of the inner tubular **204** (e.g., production tubing) to fill the volume removed by making the orbital cut **244** and fill at least a portion of the upper central bore portion **320**. As set forth above, control lines **200** may provide leak paths for hydrocarbons to travel through the seal formed in the wellbore **108** as part of the P&A operations. Accordingly, filling the orbital cut **244** with the additional sealing material **236** may fluidly isolate the lower portion **324** of the severed control line **200** from the upper portion **326** of the severed control line **200**, which may block a leak path for hydrocarbons.

FIGS. 4A-4F illustrate respective cross-sectional views of another well abandonment system **100** configured to sever control lines **200** in a wellbore **108**, in accordance with some embodiments of the present disclosure. FIG. 4A discloses the wellbore **108** having the primary plug **300** disposed in the inner tubular **204** to form a fluid barrier, as well as the inner tubular **204** having a plurality of perforations **224**. The primary plug **300** may be run-in-hole and set at a target

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location in the inner tubular **204** (e.g., the production tubing) to block fluid from traveling through the inner tubular **204** at the primary plug **300**. As set forth above, the primary plug **300** may include any suitable material. In the illustrated embodiment, the primary plug **300** comprises a cylindrical shape such that a radially outer surface **304** of the primary plug **300** may seal against a radially inner surface **208** of the inner tubular **204** to restrain fluid. However, the primary plug **300** may include any suitable shape for blocking fluid. Further, the primary plug **300** may be expandable. That is, the primary plug **300** may be run-in-hole to the target location and then expand at the target location to seal against the inner tubular **204**.

Further, the annulus **220** may be formed between the inner tubular **204** and the production casing **214** and/or the wellbore wall **122** may be left open (e.g., not filled with sealing material **236**) such that fluid may flow through the annulus **220**. In the illustrated embodiment, the annulus **220** may be sealed via a packer assembly **222** or any suitable sealing feature to restrain fluid flow from the bottom of the wellbore **108** to the surface. Further, as illustrated, the primary plug **300** may be run-in-hole to a location disposed proximate the packer assembly **222**. The packer assembly **222** and the primary plug **300**, in combination, may provide a fluid barrier between the surface and the bottom of the wellbore **108** and/or a production zone.

Moreover, with the primary plug **300** set in the inner tubular **204**, the perforating device **226** (e.g., perforating gun, blade, reamer, hydro jet, plasma torch, etc.) may be lowered into the wellbore **108** to perforate the inner tubular **204** above the primary plug **300**. In particular, the perforating device **226** may be configured to perforate the inner tubular **204** to form a first perforation zone **400** and a second perforation zone **402**. The first perforation zone **400** may be disposed proximate the primary plug **300**. In particular, the first perforation zone **400** may be disposed immediately uphole from the primary plug **300**, and the second perforation zone **402** may be disposed uphole from the first perforation zone **400**. The first perforation zone **400** and the second perforation zone **402** may be spaced apart axially (i.e., axially offset). As illustrated, a non-perforated zone **404** may be disposed between the first perforation zone **400** and the second perforation zone **402** to separate the first perforation zone **400** from the second perforation zone **402**. The non-perforated zone **404** may have an axial length between 50-200 meters, which may provide sufficient space for the secondary plug **302** (shown in FIG. 4B) discussed below.

Moreover, the first perforation zone **400** may include at least one first perforation **406**, which opens at least one first fluid path **408** between the central bore **230** of the inner tubular **204** and the annulus **220** such that the sealing material **236** may be injected into the annulus **220** from the central bore **230** via the at least one fluid path **228**. Further, as illustrated, the first perforation zone **400** may include a plurality of first perforations **406** that open a plurality of first fluid paths **408** between the central bore **230** of the inner tubular **204** and the annulus **220**. The first perforation zone **400** may extend 1.0 to 10.0 meters along the inner tubular **204**. Further, each first fluid path of the plurality of first fluid paths **408** in the first perforation zone **400** may be axially offset from at least one adjacent first fluid path along the inner tubular **204** by 1.0 to 30.0 centimeters. Additionally, the second perforation zone **402** may include at least one second perforation **410**, which opens at least one second fluid path **412** between the central bore **230** of the inner tubular **204** and the annulus **220**. Similarly, the second perforation zone **402** may include a plurality of perforations

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410 that open a plurality of second fluid paths **412** between the central bore **230** of the inner tubular **204** and the annulus **220**. The second perforation zone **402** may extend 1.0 to 10.0 meters along the inner tubular **204**. Each second fluid path of the plurality of second fluid paths **412** in the second perforation zone may be axially offset from at least one adjacent second fluid path along the inner tubular **204** by 1.0 to 30.0 centimeters.

FIG. 4B discloses a secondary plug **302** disposed in the inner tubular **204**. As illustrated, the secondary plug **302** may be run-in-hole and set in a position uphole from the primary plug **300**. In particular, the secondary plug **302** may be run-in-hole and set in a position between the first perforation zone **400** and the second perforation zone **402** such that the secondary plug **302** is disposed between the at least one first fluid path **408** and the at least one second fluid path **412**. Moreover, the secondary plug **302** may comprise the through bore **306** extending from the uphole end **308** to the downhole end **310** of the secondary plug **302** such that fluid may pass through the secondary plug **302**. However, the secondary plug **302** may also include the check-valve **312** configured to restrain fluid flow in the uphole direction **314**. The check-valve **312** may include any suitable type of check valve for restraining flow of the sealing material **236** in the uphole direction **314** and permitting fluid flow of the sealing material **236** in the downhole direction **316** through the check-valve **312**. For example, the check-valve **312** may include a flapper check valve having a flapper configured to swing open in response to fluid (e.g., the sealing material **236**) flowing in the downhole direction **316** and close in response to the fluid flowing in the uphole direction **314**.

FIG. 4C discloses the sealing material **236** disposed in the wellbore **108**. As illustrated, the sealing material **236** (e.g., cement) may be pumped through the check-valve **312** into the lower central bore portion **318** (e.g., the portion of the central bore **230** disposed between the primary plug **300** and the secondary plug **302**). The sealing material **236** may be injected through the secondary plug **302** via the injection conveyance **238** (e.g., drill pipe, coiled tubing, or some combination thereof). Moreover, the sealing material **236** may be pumped into the lower central bore portion **318** in a partially liquid state. Accordingly, as the lower central bore portion **318** is filled with the sealing material **236**, fluid pressure from the sealing material **236** entering through the check-valve **312** may drive the sealing material **236** through the at least one first fluid path **408** and into the annulus **220**. The sealing material **236** may continue to be pumped through the check-valve **312** to drive the sealing material **236** to fill a portion of the annulus **220** proximate the packer assembly **222** and continue to flow upward in the annulus **220** to a portion of the annulus **220** adjacent the second perforation zone **402**. As the sealing material **236** continues to be pumped through the check-valve **312**, the sealing material **236** may flow into the central bore **230** of the inner tubular **204**, above the secondary plug **302**, via the at least one second fluid path **412**. Moreover, as the at least one second fluid path **412** is disposed above the secondary plug **302**, the sealing material **236** pumped into and through the annulus may surround at least a portion of a control line **200** disposed uphole from the secondary plug **302**.

In some embodiments, an upper end of the annulus **220** may be sealed. For example, the wellbore **108** may be formed as part of an underwater drilling operation such that the upper end of the annulus **220** may be sealed. As such, the at least one second fluid path **412** may allow the sealing material **236** to flow into and through the annulus **220** without causing a significant pressure increases in the annu-

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lus 220 (i.e., significant pressure increases in the annulus may rupture the tubulars). That is, the at least one second fluid path 412 may provide a pressure outlet for the annulus 220.

Further, as set forth above, the sealing material 236 may include cement, plastic, resin, or some combination thereof. As set forth above, the sealing material 236 may be pumped into the lower central bore portion 318 and the annulus 220 in a partially liquid state. However, the sealing material 236 is configured to solidify after being injected into the annulus 220. For example, the sealing material 236 may include cement 118 that is mixed with water prior to being injected into the wellbore 108 such that the cement 118 is in a liquid or fluid state. However, due to the chemical reaction (e.g., hydration) between the water and the cement 118, the cement 118 may solidify or harden over time. Once hardened, the cement 118 disposed in the annulus 220 may restrain movement of the control line 200. Specifically, the hardened cement 118 may restrain lateral movement of at least a portion of the control line 200 disposed uphole from the secondary plug 302. However, as set forth in greater detail below, the injection conveyance 238 should be removed before the sealing material 236 hardens to prevent the sealing material 236 in the central bore 230 above the secondary plug 302 from trapping the injection conveyance 238 in the wellbore 108.

FIG. 4D discloses the injection conveyance 238 (shown in FIG. 4C) removed from the central bore 230 of the inner tubular 204. Once the annulus 220 is at least sufficiently filled with the sealing material 236 such that the sealing material 236 surrounds at least a portion of the control line 200 disposed uphole from the secondary plug 302, the injection conveyance 238 may be disconnected from the secondary plug 302 and pulled out-of-hole through the sealing material 236 disposed in the central bore 230 above the secondary plug 302. As such, the injection conveyance 238 must be removed before the sealing material 236 sets and is still in an at least partially liquid state. Further, prior to the sealing material 236 setting, high pressure fluid (e.g., cleaning fluid 322) may be run into the inner tubular 204 to remove the sealing material 236 disposed uphole from the secondary plug 302. As set forth above, maintaining the upper central bore portion 320 clear of sealing material 236 may allow the cutting device 246 (shown in FIG. 4D) to be lowered into position for making the orbital cut 244 (shown in FIG. 4D) without having to mill out the upper central bore portion 320. Running the high pressure fluid (e.g., the cleaning fluid 322) into the central bore 230 of the inner tubular 204 may remove at least a portion of the sealing material 236 that may have entered the upper central bore portion 320 during injection of the sealing material 236 into the wellbore 108.

FIG. 4E discloses the control line 200 and/or flat pack severed via making the orbital cut 244 at a position uphole the secondary plug 302. As illustrated, the cutting device 246 may be run-in-hole to a target location in the central bore 230 for making the at least one orbital cut 244 (e.g., a location that is axially aligned with a portion of the annulus 220 that is filled with the hardened sealing material 236). As set forth above, the sealing material 236 is pumped into the annulus 220 to a position that is uphole from the secondary plug 302. Further, as set forth above, the upper central bore portion 320 (e.g., the portion of the central bore 230 disposed above the secondary plug 302) may be substantially clear of sealing material 236. Accordingly, once the injection conveyance 238 is pulled out-of-hole and the upper central bore portion 320 is cleaned, the cutting device 246 may

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run-in-hole to the target location above the secondary plug 302 without having to mill out the upper central bore portion 320. The cutting device 246 may be run-in-hole via a cutting device conveyance 250 (e.g., a slickline, a wireline, coiled tubing, drill pipe, etc.). Further, the cutting device 246 may include any suitable cutting device. For example, the cutting device 246 may include an abrasive jetting device, a blade cutter, a reamer, a plasma torch, or some combination thereof.

Moreover, at the target location above the secondary plug 302, the cutting device 246 may be configured to cut through the inner tubular 204 (e.g., production tubing) and the sealing material 236 in the annulus 220 and sever the control line 200 surrounded by the sealing material 236. Indeed, the hardened sealing material 236 is configured to restrain at least lateral movement of the control line 200. As set forth above, the cutting device 246 may fail to cut the control line 200 if the control line is permitted to move in response to contact with the cutting device 246. However, in the illustrated embodiment, as the cutting device 246 cuts through the sealing material 236 and engages the control line 200 and/or flat pack, the remaining sealing material 236 may prevent the control line 200 and/or flat pack from moving laterally in response to contact with the cutting device 246. That is, as the cutting device 246 engages a first side of the control line 200, an opposite side of the control line 200 may interface with the remaining hardened sealing material 236 to prevent the control line 200 from moving with respect to the cutting device 246 such that the cutting device 246 may engage and sever the control line 200.

FIG. 4F discloses the sealing material 236 disposed in the orbital cut 244 and the upper central bore portion 320. After making the orbital cut 244, additional sealing material 236 may be injected into the upper central bore portion 320 of the inner tubular 204 (e.g., production tubing) to fill the volume removed by making the orbital cut 244 and fill at least a portion of the upper central bore portion 320. As set forth above, the control lines 200 may provide leak paths for hydrocarbons to travel through a seal formed in the wellbore 108 as part of the P&A operations. Accordingly, filling the orbital cut 244 with the additional sealing material 236 may fluidly isolate the lower portion 324 of the severed control line 200 from the upper portion 326 of the severed control line 200, which may block a leak path for hydrocarbons.

Accordingly, the present disclosure may systems and methods for severing control lines and/or flat packs during plug and abandonment (P&A) operations. The systems and methods may include any of the various features disclosed herein, including one or more of the following statements.

Statement 1. A method, comprising: injecting a sealing material into an annulus formed between a tubular and a wellbore wall, wherein the sealing material is configured to surround at least a portion of a control line disposed in the annulus; making at least one orbital cut, via a cutting device, through the tubular, the sealing material in the annulus, and the control line surrounded by the sealing material; and filling the at least one orbital cut and at least a portion of a central bore of the tubular with the sealing material to seal a wellbore.

Statement 2. The system of statement 1, wherein the cutting device comprises an abrasive jetting device, a blade cutter, a reamer, a plasma torch, or some combination thereof.

Statement 3. The system of any preceding statement, further comprising running a plug into a wellbore, via a plug conveyance, to a sealing position in a tubular disposed in the

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wellbore, wherein the plug conveyance comprises a slickline, a wireline, coiled tubing, drill pipe, or some combination thereof.

Statement 4. The system of any preceding statement, further comprising perforating a tubular above a plug disposed in a wellbore, via a perforating device, to open at least one fluid path between the central bore of the tubular and the annulus formed between the tubular and the wellbore wall.

Statement 5. The system of any preceding statement, wherein the perforating device comprises a perforating gun having shaped charges configured to detonate in a substantially radially outward direction to perforate the tubular above the plug disposed in the wellbore.

Statement 6. The system of statements 1-4, wherein the perforating device comprises the cutting device, and wherein the cutting device is configured to perforate the tubular above the plug disposed in the wellbore.

Statement 7. The system of any preceding statement, wherein the sealing material is injected into the annulus from the central bore of the tubular via the at least one fluid path extending from the central bore of the tubular to the annulus.

Statement 8. The system of any preceding statement, wherein the sealing material is injected into the annulus in at least a partially liquid state, and wherein the sealing material is configured to solidify in the annulus to restrain lateral movement of the control line.

Statement 9. The system of any preceding statement, wherein the sealing material comprises cement, plastic, resin, or some combination thereof.

Statement 10. The system of any preceding statement, further comprising milling out at least a portion of the sealing material in the central bore of the tubular, via a milling device, before making the at least one orbital cut.

Statement 11. The system of any preceding statement, wherein the milling device is run-in-hole via a drilling bit conveyance, wherein the milling device conveyance comprises a drill pipe, coiled tubing, or some combination thereof, and wherein a mud motor is configured to drive rotation of the milling device during operation.

Statement 12. The system of any preceding statement, wherein the cutting device is run-in-hole to a milled out portion of the central bore of the tubular to make the at least one orbital cut.

Statement 13. The system of any preceding statement, wherein the cutting device is run-in-hole via a cutting device conveyance, wherein the cutting device conveyance comprises a slickline, a wireline, coiled tubing, drill pipe, or some combination thereof.

Statement 14. A method, comprising: running a primary plug into a wellbore to a sealing position in a tubular disposed in the wellbore; perforating the tubular above the primary plug, via a perforating gun, to open at least one fluid path between a central bore of the tubular and an annulus formed between the tubular and a wellbore wall; running a secondary plug into the wellbore to a position above the at least one fluid path; injecting cement through the secondary plug and into the annulus via the at least one fluid path from the central bore of the tubular, wherein the cement is configured flow into the annulus to a position uphole from the secondary plug in the central bore to surround at least a portion of a control line disposed uphole from the secondary plug; making at least one orbital cut at a position uphole the secondary plug, via a cutting device, through the tubular, the cement in the annulus, and the control line surrounded by the cement; and injecting cement into the wellbore to fill areas removed by the cutting device to seal the wellbore.

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Statement 15. The system of statement 14, wherein the secondary plug comprises a check-valve configured to restrain fluid flow in an uphole direction.

Statement 16. The system of statement 14 or statement 15, wherein the cement is pushed downhole through the secondary plug via a dart driven downhole with fluid pressure from a cleaning fluid.

Statement 17. The system of statement 14 or statement 15, wherein the cement is injected through the secondary plug via an injection conveyance, wherein the injection conveyance comprises drill pipe, coiled tubing, or some combination thereof.

Statement 18. The system of statements 14-17, further comprising pulling the injection conveyance out-of-hole and running high pressure fluid into the tubular to remove the cement disposed uphole from the secondary plug.

Statement 19. A method, comprising: running a primary plug into a wellbore to a sealing position in a tubular disposed in the wellbore; perforating the tubular above the primary plug, via a perforating gun, to open a first fluid path and a second fluid path each extending between a central bore of the tubular and an annulus formed between the tubular and a wellbore wall, wherein the first fluid path and the second fluid path are axially offset from each other along the wellbore; running a secondary plug into the wellbore to a position between the first fluid path and the second fluid path; injecting cement, via a conveyance, through the secondary plug and into the annulus via the first fluid path from the central bore of the tubular, wherein the cement is configured flow into the annulus to a position uphole from the secondary plug in the central bore to surround at least a portion of a control line disposed uphole from the secondary plug, and wherein a portion of the cement is configured to flow from the annulus into the central bore via the second fluid path; removing the conveyance from the wellbore; circulating fluid in the central bore to remove the cement disposed in the central bore above the secondary plug; making at least one orbital cut at a position uphole the secondary plug, via a cutting device, through the tubular, the cement in the annulus, and the control line surrounded by the cement; and injecting cement into the wellbore to fill areas removed by the cutting device to seal the wellbore.

Statement 20. The system of statement 19, wherein the perforating gun is configured to perforate the tubular to form a first perforation zone comprising the first fluid path and to form a second perforation zone comprising the second fluid path, and wherein a non-perforated zone is disposed between the first perforation zone and the second perforation zone.

For the sake of brevity, only certain ranges are explicitly disclosed herein. However, ranges from any lower limit may be combined with any upper limit to recite a range not explicitly recited, as well as, ranges from any lower limit may be combined with any other lower limit to recite a range not explicitly recited, in the same way, ranges from any upper limit may be combined with any other upper limit to recite a range not explicitly recited. Additionally, whenever a numerical range with a lower limit and an upper limit is disclosed, any number and any included range falling within the range are specifically disclosed. In particular, every range of values (of the form, "from about a to about b," or, equivalently, "from approximately a to b," or, equivalently, "from approximately a-b") disclosed herein is to be understood to set forth every number and range encompassed within the broader range of values even if not explicitly recited. Thus, every point or individual value may serve as its own lower or upper limit combined with any other point

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or individual value or any other lower or upper limit, to recite a range not explicitly recited.

Therefore, the present embodiments are well adapted to attain the ends and advantages mentioned as well as those that are inherent therein. The particular embodiments disclosed above are illustrative only, as the present embodiments may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. Although individual embodiments are discussed, all combinations of each embodiment are contemplated and covered by the disclosure. Furthermore, no limitations are intended to the details of construction or design herein shown, other than as described in the claims below. Also, the terms in the claims have their plain, ordinary meaning unless otherwise explicitly and clearly defined by the patentee. It is therefore evident that the particular illustrative embodiments disclosed above may be altered or modified and all such variations are considered within the scope and spirit of the present disclosure.

What is claimed is:

1. A method, comprising:

performing a tubular above a plug disposed in the tubular, via a perforating device, to open at least one fluid between a central bore of the tubular and an annulus formed between the tubular and a wellbore wall of a wellbore, wherein the plug is configured to block fluid flow through the tubular;

injecting a sealing material into the central bore of the tubular and into the annulus formed between the tubular and the wellbore wall via the sealing material flowing through the at least one fluid path from the central bore, wherein the sealing material is configured to surround at least a portion of a control line disposed in the annulus, and wherein the sealing material is configured to fill the central bore to a same height or higher than the annulus;

milling out at least a portion of the sealing material disposed in the central bore above the plug, via a milling device, to a height lower than the sealing material in the annulus;

lowering a cutting device into a milled out portion of the central bore to position that is axially aligned with a portion of the annulus having the sealing material;

making at least one orbital cut, via the cutting device, through the tubular, the sealing material in the annulus, and the control line surrounded by the sealing material; and

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filling the at least one orbital cut and at least a portion of the central bore of the tubular with the sealing material to seal the wellbore.

2. The method of claim 1, wherein the cutting device comprises an abrasive jetting device, a blade cutter, a reamer, a plasma torch, or some combination thereof.

3. The method of claim 1, further comprising running the plug into the wellbore, via a plug conveyance, to a sealing position in the central bore of the tubular disposed in the wellbore, wherein the plug conveyance comprises a slickline, a wireline, coiled tubing, drill pipe, or some combination thereof.

4. The method of claim 1, wherein the perforating device comprises a perforating gun having shaped charges configured to detonate in a substantially radially outward direction to perforate the tubular above the plug disposed in the wellbore.

5. The method of claim 1, wherein the perforating device comprises the cutting device, and wherein the cutting device is configured to perforate the tubular above the plug disposed in the wellbore.

6. The method of claim 1, wherein the sealing material is injected into the annulus in at least a partially liquid state, and wherein the sealing material is configured to solidify in the annulus to restrain lateral movement of the control line.

7. The method of claim 1, wherein the sealing material comprises cement, plastic, resin, or some combination thereof.

8. The method of claim 1, wherein the milling device is run-in-hole via a drilling bit conveyance, wherein the milling device conveyance comprises a drill pipe, coiled tubing, or some combination thereof, and wherein a mud motor is configured to drive rotation of the milling device during operation.

9. The method of claim 1, wherein the cutting device is run-in-hole via a cutting device conveyance, wherein the cutting device conveyance comprises a slickline, a wireline, coiled tubing, drill pipe, or some combination thereof.

10. The method of claim 1, further comprising injecting cleaning fluid into the central bore of the tubular to clean the central bore, the annulus, or some combination thereof, after perforating the tubular.

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